Project 3: Active Learning for Learning-to-Defer
Human-Centric Artificial Intelligence

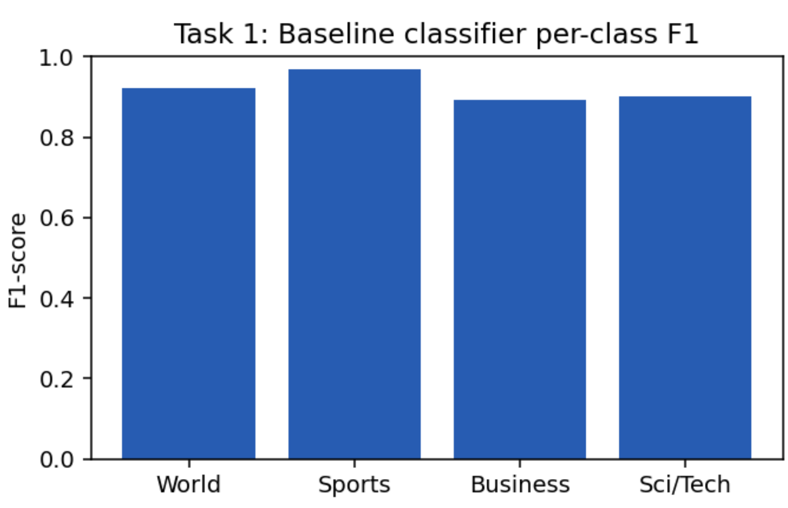
This report describes the design and results of a human-AI collaboration system built on the AG News topic classification task. The system can either classify a news article itself or defer the decision to a simulated human expert. We progress through four stages: (1) a supervised baseline classifier, (2) a simulated expert with realistic, uneven competence, (3) a learned deferral policy trained with full access to expert labels, and (4) an active learning strategy that learns the same policy from a small budget of expert queries instead.

1. Baseline Classifier

AG News is a 4-class topic classification dataset (World, Sports, Business, Sci/Tech) with 120,000 training articles and 7,600 test articles, evenly split across classes. We use a TF-IDF vectorizer (unigrams + bigrams, 50,000 features, English stop-words removed) feeding a multinomial Logistic Regression classifier. This combination was chosen deliberately over a deep model: it trains in seconds rather than the many minutes a fine-tuned transformer would require, which matters because the later tasks retrain or re-evaluate this classifier's predictions many times over (once per query-budget point in Task 4). TF-IDF + Logistic Regression is also a strong, well-established baseline for topic classification, so accuracy is not meaningfully sacrificed for that speed.

Test accuracy: 0.9217

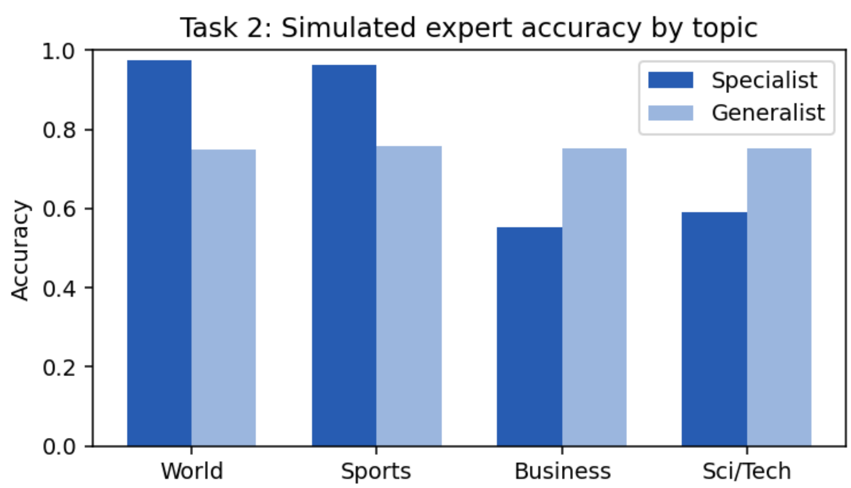
Class	Precision	Recall	F1-score	Support
World	0.932	0.911	0.921	1900
Sports	0.957	0.983	0.970	1900
Business	0.896	0.889	0.893	1900
Sci/Tech	0.900	0.904	0.902	1900



2. Simulated Expert

Real human experts are not uniformly reliable across every topic: someone who reads a lot of world and sports news will label those articles almost perfectly, but will be far shakier on business or technical jargon. We model this with a **class-conditional noisy oracle** (the "Specialist"): given the true label, it returns the correct label with a class-specific probability, and otherwise returns a uniformly random wrong label. The chosen probabilities are 0.97 (World), 0.96 (Sports), 0.55 (Business), and 0.60 (Sci/Tech) -- deliberately uneven, so that later tasks have a genuine, structured competence profile to discover rather than a single flat accuracy number. As a control, we also implement a "Generalist" expert with a single, uniform accuracy across all four classes, which lets us later show that a *targeted* active learning strategy only pays off against an expert whose competence actually varies by region of the input space.

Expert	Overall	World	Sports	Business	Sci/Tech
Specialist	0.770	0.975	0.963	0.552	0.592
Generalist	0.752	0.750	0.758	0.752	0.751



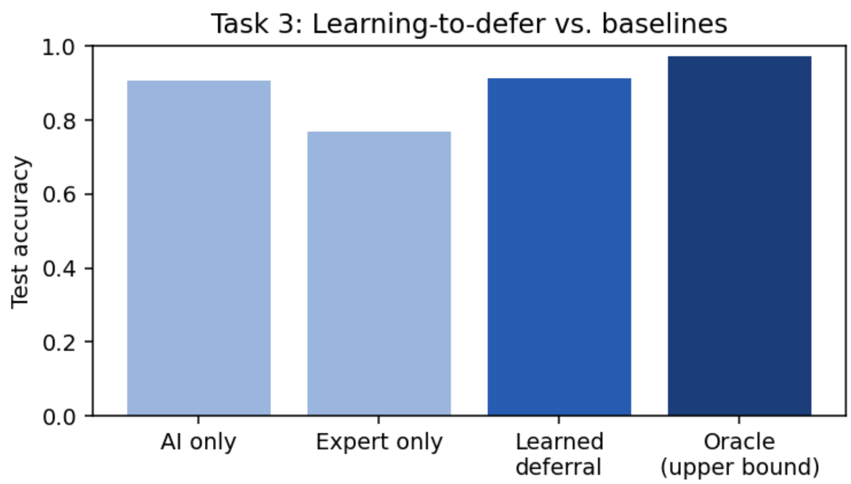
As designed, the Specialist is nearly perfect on World/Sports and considerably weaker on Business/Sci-Tech, while the Generalist is flat at roughly 0.75 everywhere. This is the "expertise profile" the deferral system in Task 3 and the active learning strategy in Task 4 need to exploit.

3. Learning to Defer

With both the AI classifier's predictions and (simulated) expert labels available during training, we implement a deferral policy as a small, separately-trained **logistic-regression gate**, rather than a fixed confidence threshold. Three features are computed from the AI classifier's predicted probability distribution for each example: its top-1 confidence, its margin (top-1 minus top-2 probability), and its entropy. The gate is trained on a held-out slice of the training set to predict a binary target: *defer = 1* exactly when the AI's prediction is wrong AND the expert's prediction is right -- i.e., when deferring would actually flip a mistake into a correct answer. This directly targets combined system accuracy instead of only the AI's own calibration.

Because "deferring helps" is a rare class, the gate's raw probability output needs a carefully chosen operating threshold rather than the default 0.5 cutoff: at 0.5 the gate over-defers and can even underperform the AI-only baseline. We instead calibrate the threshold on a further held-out slice by directly maximising system accuracy, which raised the optimal cutoff to 0.85 in our experiments and made deferral much more selective (a deferral rate around 8%, versus over 20% at the uncalibrated default).

Metric	Value
AI only	0.9068
Expert only	0.7703
Learned deferral system	0.9139
Oracle upper bound	0.9722
Deferral rate	0.083
Accuracy on deferred subset	0.649
Accuracy on kept subset	0.938
Calibrated gate threshold	0.850

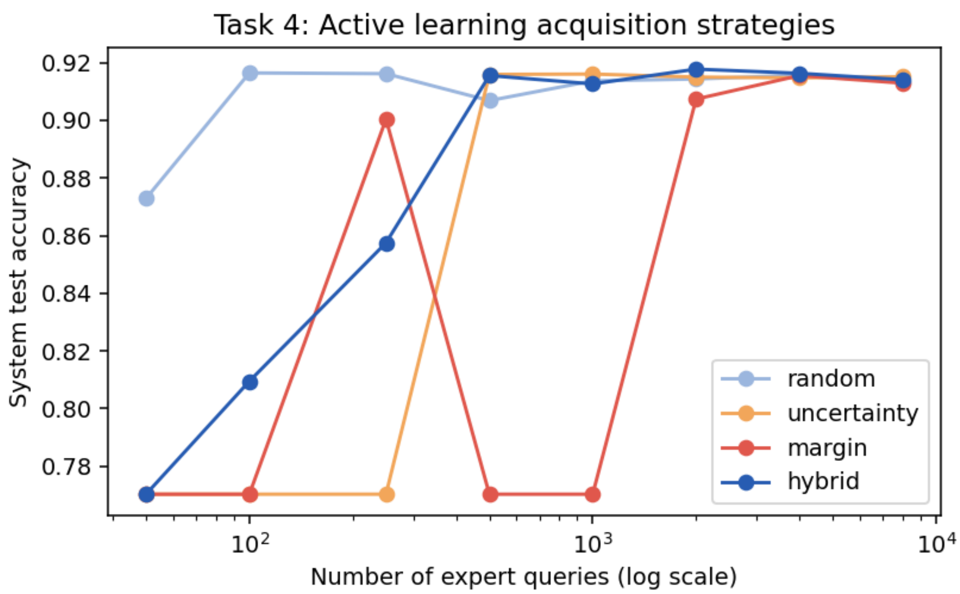


The learned deferral system reaches 91.4% accuracy, above the AI-only baseline (90.7%) and well above the expert-only baseline (77.0%), while only consulting the expert on 8.3% of test examples. It recovers a meaningful fraction of the gap to the oracle upper bound (97.2%) -- the best any deferral rule could achieve with this particular AI and expert. Notably, accuracy on the deferred subset (64.9%) is well below the expert's overall accuracy: this is expected, since deferral is targeted precisely at the AI's hardest, most ambiguous examples, where the expert also struggles -- just slightly less than the AI does.

4. Active Learning for Expert-Competence Discovery

We now assume no expert labels are available up front, and that each expert query has a cost, so the goal is to learn the same deferral policy as in Task 3 using as few queries as possible. We compare four acquisition strategies for selecting which training points to query: **random** sampling (control), **uncertainty** sampling (query the AI's least confident points first), **margin** sampling (query the smallest gap between the AI's top-2 class probabilities), and a **hybrid** strategy that queries half its budget randomly and half by uncertainty. For each budget we select that many points, query the simulated expert only on them, train a Task-3-style deferral gate using only that labelled subset (with a 70/30 fit/calibration split when the budget allows it), and evaluate the resulting system on the full held-out test set.

Uncertainty and margin sampling are natural choices here because the deferral decision is fundamentally about the AI's own confidence signal: concentrating the query budget near the AI's decision boundary should reveal the most information about when deferral helps, for the fewest queries.



Queries	random	uncertainty	margin	hybrid
50	0.8729	0.7703	0.7703	0.7703
100	0.9164	0.7703	0.7703	0.8095
250	0.9162	0.7703	0.9004	0.8575
500	0.9068	0.9159	0.7703	0.9155
1000	0.9137	0.9161	0.7703	0.9126
2000	0.9143	0.9150	0.9074	0.9178
4000	0.9157	0.9147	0.9155	0.9163
8000	0.9133	0.9151	0.9129	0.9141

A striking and, in hindsight, explainable failure mode appears at very small query budgets (50-250 queries): pure uncertainty and margin sampling can produce a **degenerate "always defer" gate**, collapsing system accuracy to exactly the expert-only accuracy (0.7703). Because these strategies only ever query the AI's hardest, least confident points, the queried sample can end up containing almost no examples where the AI was actually correct -- so the gate never learns the contrast needed to say "don't defer here." The hybrid strategy, which reserves half its budget for random exploration, avoids this collapse and reaches a reasonable accuracy (0.81-0.86) even at the smallest budgets, though it is not always the single best strategy once the budget grows large enough that pure uncertainty/margin sampling stabilises (from roughly 500 queries onward, all strategies except the smallest random-only draws converge to within about one percentage point of the fully-supervised Task 3 result). This suggests a practical recommendation for deploying such a system with a real expert: start with a hybrid acquisition strategy while the query budget is small, since it is markedly more robust to the exact points drawn, and pure uncertainty sampling only becomes reliable once a few hundred queries have been collected.

5. Optional Extension and Limitations

As an optional extension, the project interface includes a small interactive page where a visitor can paste an arbitrary news snippet and see the baseline classifier's prediction, its confidence, and the calibrated Task 3 threshold's deferral decision in real time. This is a lightweight stand-in for a full expert-in-the-loop interface, rather than a complete implementation of one.

The main limitation of this project is that the expert is entirely simulated: its class-conditional accuracy profile is hand-specified rather than measured from a real person, so the active learning results in Task 4 should be read as a demonstration of the acquisition strategies' behaviour under a known, controlled competence profile, rather than a claim about how quickly a real deployment would converge. A natural next step would be to replace the simulated expert with the interactive interface and repeat the Task 4 comparison against a genuine, unknown competence profile.